

SNIGDHA GAYATHRI MANDAPATI

AI Engineer | LLM Systems, RAG Pipelines & Multi-Agent Orchestration

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PROFESSIONAL SUMMARY

AI Engineer specializing in **LLM-powered systems**, **agentic architectures**, and production-grade **RAG pipelines**, with hands-on depth across the full GenAI stack: **transformer** internals, **retrieval and vector search**, **fine-tuning** (LoRA/QLoRA), multi-agent orchestration with **LangChain/LangGraph**, and **MLOps** for model serving and evaluation. Proven track record shipping end-to-end AI systems: from **5-agent orchestration logic** and security-hardened RAG pipelines to **privacy-preserving hybrid ML architectures**. Combines strong software engineering fundamentals (Python, FastAPI, React, PostgreSQL) with a security- and evaluation-first mindset, consistently **reducing latency**, **hallucination risk**, and **model drift in deployed systems**.

SKILLS

LLM & Generative AI Engineering: Large Language Models, Transformer Architecture, Prompt Engineering, LLM Evaluation & Benchmarking, Hallucination Mitigation, Generative AI, Context Engineering, Gemini API, Hugging Face Transformers

Agentic AI & Orchestration: LangChain, LangGraph, Multi-Agent Systems, Agent Memory, Tool Use / Function Calling, Model Context Protocol (MCP), Structured Output Generation, Agent Evaluation & Reliability Patterns

RAG & Retrieval: Retrieval-Augmented Generation, Vector Databases, Embedding Models (SentenceTransformers), Semantic Search, Chunking Strategies, Reranking, Grounding & Faithfulness Verification, RAGAS

Fine-Tuning & Model Development: LoRA / QLoRA, Supervised Fine-Tuning (SFT), Scikit-Learn, XGBoost, TensorFlow, PyTorch, Quantum ML (PennyLane), Model Evaluation (F1, ROC-AUC, RMSE)

ML, MLOps & Deployment: Machine Learning, Deep Learning, NLP, Data pre-processing, Data Analysis, ML pipelines, Docker, FastAPI, Flask REST APIs, CI/CD, Model Monitoring & Drift Detection, LLM Observability (LangSmith), Cost & Latency Optimization

Programming & Data: Python, JavaScript/TypeScript, SQL, Java, C++, PostgreSQL, MySQL, Next.js, React, Git, JWT/Auth (Clerk)

TECHNICAL EXPERIENCE

Machine Learning Intern | Cognifyz Technologies

Jan 2025 – Mar 2025

- **Deployed** 10+ supervised ML models (Scikit-Learn, XGBoost) as Dockerized Flask REST APIs, **sustaining sub-200ms latency** across **10K+ daily requests in production**.
- **Reduced** production **model drift by ~30%** by designing and shipping automated monitoring and retraining pipelines, while achieving **95% validation accuracy** on skewed real-world datasets.

Artificial Intelligence Intern | Technical Hub Pvt. Ltd.

May 2024 – Sep 2024

- **Increased retrieval precision by 40% over the prior baseline** by architecting multi-stage classification pipelines (PCA, ensemble methods) with iterative feature engineering.
- **Cut inference pipeline processing time by 35%** and **accelerated model iteration cycles by 25%** by profiling and re-engineering the ML pipeline, shortening the team's experiment-to-result loop.

PROJECTS

NeuroPlan AI | Multi-Agent Adaptive Learning System

[Link](#)

- **Architected a 5-agent orchestration system** (Roadmap Creation, Fatigue Monitoring, Transfer Network, Mastery Radar, Planner) that autonomously generates, monitors, and re-optimizes personalized learning roadmaps in real time.
- **Built** cross-agent coordination logic enabling the Planner Agent to arbitrate fatigue, mastery, and transfer signals and dynamically re-sequence tasks, on a FastAPI + Next.js + PostgreSQL stack with Gemini-powered generation and Clerk-secured access.

Smart Shelf AI | Hybrid Quantum-Classical Recommendation Engine

[Link](#)

- **Reduced** recommendation latency from a 1–2 minute cold start to sub-1-second on repeat queries by engineering a **hybrid classical quantum pipeline** (SentenceTransformers 384-dim embeddings + PennyLane 6-qubit similarity kernel) with a precomputed caching layer.
- **Delivered** a 100% local-inference, **zero-external-dependency recommendation engine** : including **27-class emotion detection** (GoEmotions) : served through a dedicated **FastAPI backend for full data privacy**.

Placement RAG Agent | Security-Hardened, Production-Grade RAG Pipeline

[Link](#)

- **Hardened** an LLM-powered RAG system against prompt injection and data-poisoning attacks, **achieving 90%+ test coverage** across a **9-stage security pipeline** (injection detection, retrieval guarding, grounding verification, output sanitization) built in FastAPI.
- **Cut** hallucinated responses reaching end users by enforcing a **0.45 grounding-similarity threshold** via a dedicated verification module that cross-checks generated answers against retrieved context before output.

EDUCATION

B.Tech in Computer Science & Engineering (AI/ML) — Aditya University, Surampalem

2022 – 2026 | CGPA: 8.30

CERTIFICATIONS & ACHIEVEMENTS

IBM (ML with Python, DL Fundamentals, Data Science 101) | Microsoft Azure AI Fundamentals | Google Career Essentials in Generative AI · Pearson IT Specialist – Java | Introduction to LangChain – Python, Building Reliable Agents by Langchain Academy | LangChain for LLM App Development by SimpliLearn